Video Transcript

Video 2: Relational Databases - Part 1 (04:46)

The relational database management system is the traditional database. That is, it goes back for decades and MySQL is the most widely used relational database management system. So, it's a good one to get familiar with as you will encounter it quite a bit. Now, as you can see here, the way a relational database management lays out the data model, its design of the information in the world that you're trying to capture is that it does so by creating tables and relationships between those tables. More formally, doing a database design involves choosing the tables, the columns and those relationships.

So, if we consider a simple example, say of students and colleges, you can imagine creating a table of students and their information in one of universities and colleges and their information. In database speak, we would talk about entities for the tables and the properties for the columns. Now, there are a few other things that you would include in a relational database design. And as you can see there, one of them would be the identifiers, that is the unique handle for each of those rows and then the relationships between the tables. You can imagine that if you have students and you have colleges, you'd want to have a relationship.

Now, if you want to see a fuller example, go ahead and take a look at the Sakila sample database that is included in many of the MySQL distributions, installations. And if you don't have it, you can search it, download it, and install it for your further study. Now, one of the very first things that we're going to want to do is that we're going to want to connect to that database server. And the way we're going to be doing it is by running a Python program at the command line, as you can see here, and connecting to that database. In order to do that, to write that program, if you don't already have it, go ahead and install Visual Studio Code, then install Python. And once you do that, as you can see here, in addition to the command line prompt that you see there on the laptop and the Python program or runtime that you will need, there's also a driver. That driver captures, within its package, all of the functionality that is needed for you to talk to the database. The one that we will be using, as you can see here in the one that you will need to install is MySQL-connector-Python. Now, once we have installed it, we will confirm that that is available by running a program in Python with just simply one line that simply imports that package.

So, let's go ahead and write that now. So, let's start off by opening up the command line as you can see here. Then I'm going to go ahead and paste that instruction, the same one that we had on the slides and in my installation of Python, because this operating system already has another one and I need to run the most recent one, I will add pip three to that instruction and then I'm going to go ahead and run it. And as you can see there, that package was downloaded and installed.

And now I'm going to go ahead and minimize that window and I'm going to go ahead and drag and drop the folder that I have here on the desktop. Once that opens, you will see that I have a single file here. And once I open that file, I'm going to go ahead and ignore this prompt. You'll see that it has a single line that we saw on the slides and it is simply import in that package. So, I'm going to go ahead and go back to the command line and I'm going to go ahead and navigate to this folder.

So, I'm going to go ahead and enter here desktop, then sample, I'm going to go ahead and give it permission to access my desktop. And now I'm going to go ahead and list my files to make sure I am in the right directory. And I'm going to go ahead and enter Python 3. And then I'm going to go ahead and enter the driver name. Now, if this works, that means it finds the package, then we're fine. Let's go ahead and run it and as you can see there that executed without an issue. Now, if you see an error, that means there is a problem with your program accessing your driver and you will have to double and debug that. If not, then let's go ahead and go forward. And if you had some problem, make sure you resolve it before you go on.